DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 —NOTES

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Common Network Protocols and Ports

Learning Objective

To understand some common network protocols, their default ports, and how they are used in network communication and SOC investigations.

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1. What are Network Protocols?

Network protocols are basically a set of rules that devices use to communicate with each other over a network.

Different protocols are used for different purposes.

For example, DNS is used to resolve domain names, HTTP is used for web communication, and SSH is used for remote access.

Knowing common protocols and ports is important for a SOC analyst because network traffic often contains information about what a device is doing.

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2. Common Protocols and Default Ports

Protocol	*Port*	*Purpose*
HTTP	*80*	*Used for communication between a web browser and a web server*
HTTPS	*443*	*Used for secure web communication using encryption*
DNS	*53*	*Used to translate domain names into IP addresses*
DHCP	*67/68*	*Used to automatically provide IP configuration to devices*

SSH	*22*	*Used for secure remote access to a system*
FTP	*21*	*Used for transferring files between systems*
SMTP	*25*	*Used for sending email between mail servers*

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3. HTTP

HTTP (Hypertext Transfer Protocol) is used when a browser communicates with a web server.

The default port for HTTP is:

Port 80

For example, when I open a website using HTTP, my browser sends requests to the web server and the server sends back the requested content.

HTTP traffic is not encrypted, so sensitive information should not normally be sent over plain HTTP.

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4. HTTPS

HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP.

The default port is:

Port 443

HTTPS uses encryption through TLS to protect the communication between the browser and the server.

As a SOC analyst, I may commonly see TCP connections to port 443 when investigating normal web browsing as well as potentially suspicious connections.

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5. DNS

DNS (Domain Name System) is used to convert domain names into IP addresses.

The default port is:

Port 53

DNS commonly uses UDP for normal queries, although TCP can also be used in some situations.

For example:

google.com → IP address

Instead of remembering an IP address, users can use a domain name.

DNS logs can also be useful during security investigations because suspicious domains may appear in DNS queries.

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6. DHCP

DHCP (Dynamic Host Configuration Protocol) automatically provides network configuration to devices.

It commonly uses:

- **UDP 67** — DHCP server
- **UDP 68** — DHCP client

It can provide information such as:

- IP address
- Subnet mask
- Default gateway
- DNS server

Without DHCP, network settings may have to be configured manually.

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7. SSH

SSH (Secure Shell) is used to securely connect to another system remotely.

The default port is:

Port 22

It is commonly used by administrators to manage Linux servers and other systems remotely.

From a SOC point of view, unexpected SSH connections can be worth investigating, especially if they come from an unusual IP address or involve an unusual account.

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8. FTP

FTP (File Transfer Protocol) is used to transfer files between systems.

The default control port is:

Port 21

FTP does not provide encryption by default, so usernames, passwords and data can potentially be exposed.

There are more secure alternatives such as SFTP, which works over SSH.

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9. SMTP

SMTP (Simple Mail Transfer Protocol) is mainly used for sending email.

The traditional default port is:

Port 25

SMTP is mainly used for mail server-to-mail server communication.

Other ports such as 587 are commonly used for mail submission, but the port asked for in this assignment is **25**.

For a SOC analyst, email traffic can be important when investigating phishing, spam or suspicious email activity.

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10. Why Ports Matter in SOC Investigations

Ports can give an analyst an idea about what type of communication is happening.

For example:

- Port **22** → possible SSH connection
- Port **53** → DNS traffic
- Port **80** → HTTP traffic
- Port **443** → HTTPS traffic
- Port **25** → SMTP traffic

However, I should not assume that a port automatically means a specific type of traffic is malicious.

Attackers can use unusual ports or services can be configured to use different ports.

So I would use the port as **one piece of evidence** and investigate the source, destination, protocol, timing and other related activity.

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11. My Understanding

I learned that knowing common ports and protocols is important for understanding network traffic.

For example, if I see a device communicating with port 53, I can check whether it is DNS traffic and what domains are being requested.

Similarly, if I see an unexpected connection to port 22, I can investigate whether someone is using SSH to remotely access the system.

So, ports alone don't tell me whether traffic is malicious, but they give me a useful starting point during an investigation.